



AI-POWERD CONGNITIVE SCREENING AND ANALYSIS PLATFORM

25-26J-439

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Problem Statement & Stakeholders

Context: AI-based platform for **early dementia screening and cognitive analysis**.

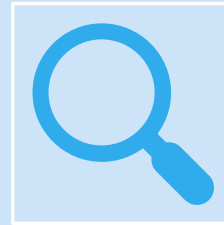
Stakeholders: Elderly individuals, Clinicians & healthcare professionals, Hospitals, caregivers

Observed issue: Over 60% of dementia cases are detected at moderate or late stages due to lack of accessible early screening tools.

Why it matters: Late detection increases **costs and caregiver burden (up to 40–50%)**, while **early detection improves treatment outcomes and patient quality of life**.



Research Gap & Objectives



Research Gap -

Most existing systems rely on single-modality analysis (speech, MRI, or cognitive tests) and lack end-to-end multimodal pipelines for remote dementia screening and AI-supported clinical diagnosis with explainable result



Main objective -

To Develop a multimodal AI-based platform that analyzes handwriting, speech, lifestyle data, and MRI scans to support early dementia screening and analysis.

Solution Overview (What You Built)

- Solution summary

A **multimodal AI-powered cognitive screening and analysis platform** that enables **remote early dementia screening** using handwriting, speech, health monitoring, and MRI analysis.



Key features delivered

- **FR1:** Handwriting-based cognitive screening using writing dynamics
- **FR2:** Speech-task analysis for cognitive risk detection
- **FR3:** Smart health monitoring and lifestyle-based dementia risk prediction
- **FR4:** MRI-based dementia stage classification with explainable AI

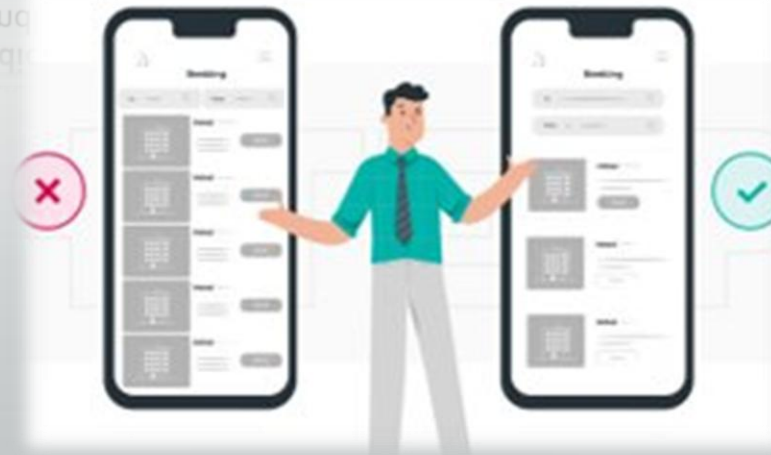
Solution Overview (What You Built)

Non-functional

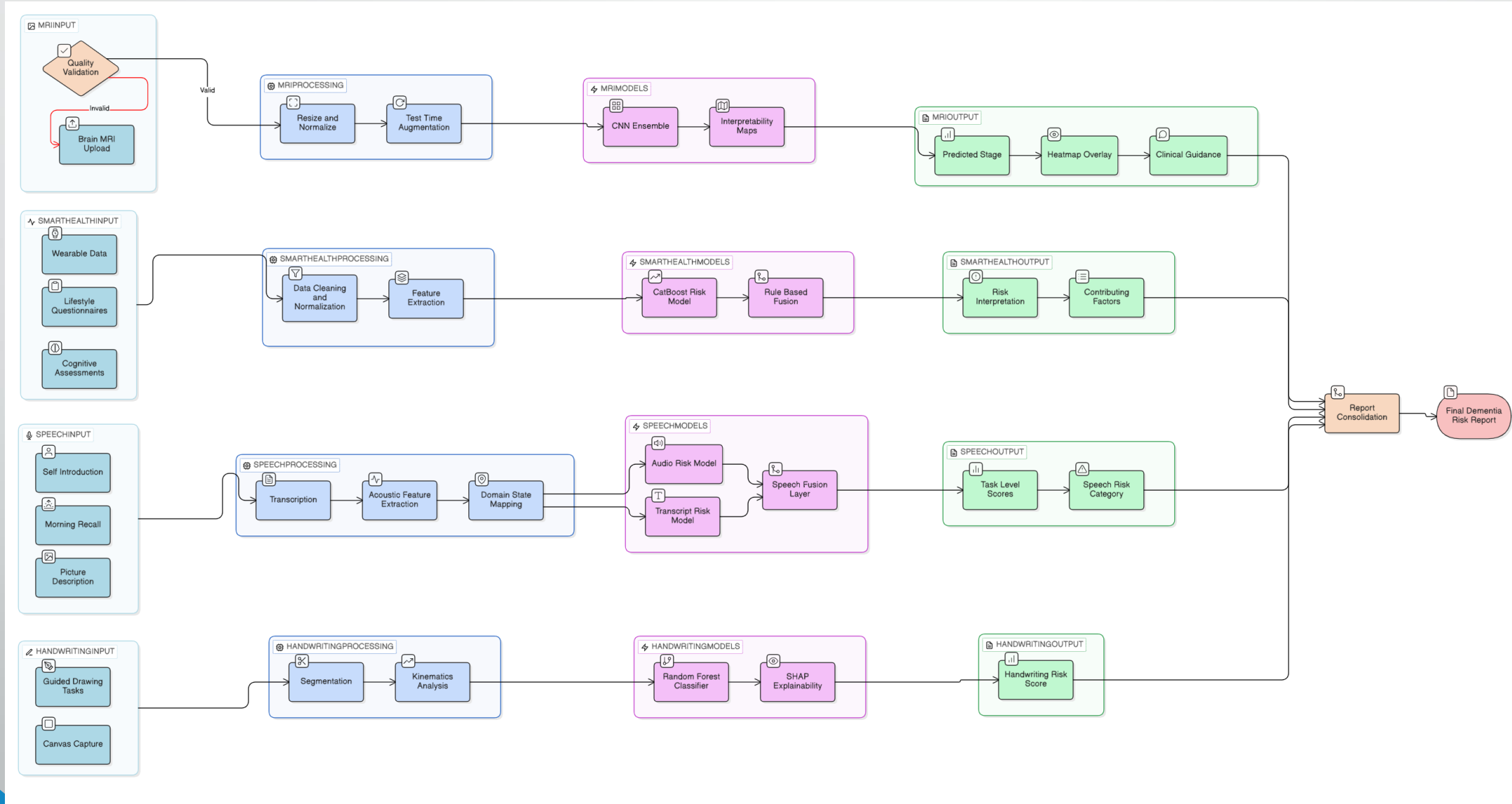
- **focus Usability** – simple browser-based screening **tasks**.
- **Reliability** – machine learning models with interpretable outputs.
- **Performance** – efficient data processing and **prediction**.
- **Security** – secure handling of patient data

Current completion status

-90% system implementation completed



System Architecture & Integration



Commercialization & Research Impact

Target Users / Market

- Adults 55+ at early cognitive risk
- Neurology & psychiatry clinics
- Primary care & rural health centers
- Government elderly screening programs (Sri Lanka → South Asia expansion)

Global burden:

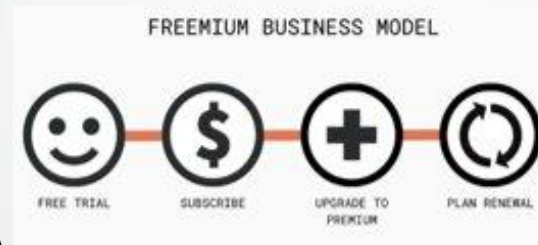
- 55+ million living with dementia (WHO)
- 10 million new cases annually
- Low-cost early screening

Value Proposition (Compared to Alternatives)

- Multimodal AI platform:
 - Speech-based model
 - Handwriting analysis
 - Wearable physiological indicators
- Non-invasive, remote-capable (<25 min screening)
- Multilingual: English, Sinhala, Tamil
- Explainable AI support for clinician trust
- Designed for low-resource settings

Cost & Feasibility

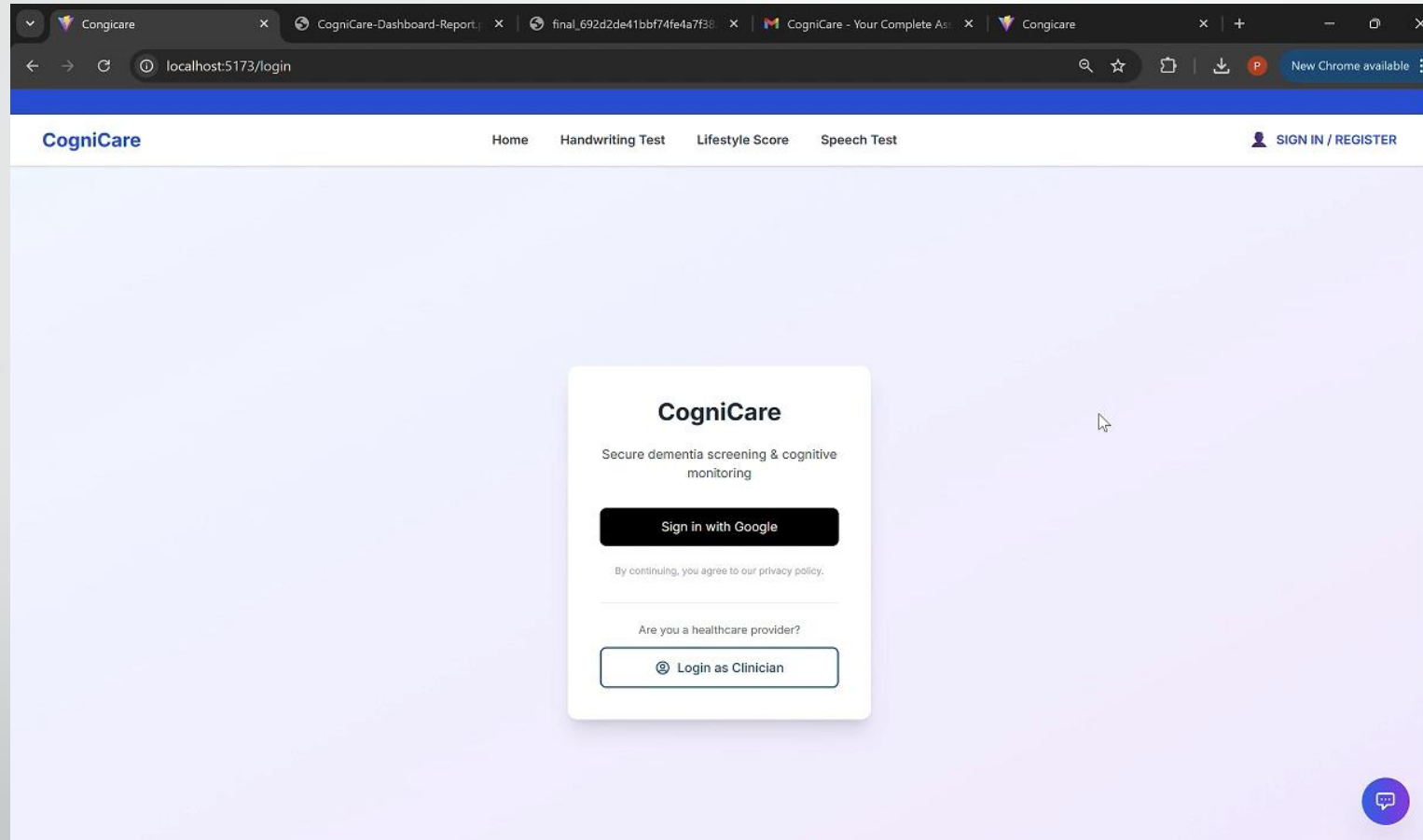
- Pretrained models reduce training cost
- Cloud-based deployment (low hardware requirement)
- Minimal per-screening operational cost
- Uses open-source AI frameworks



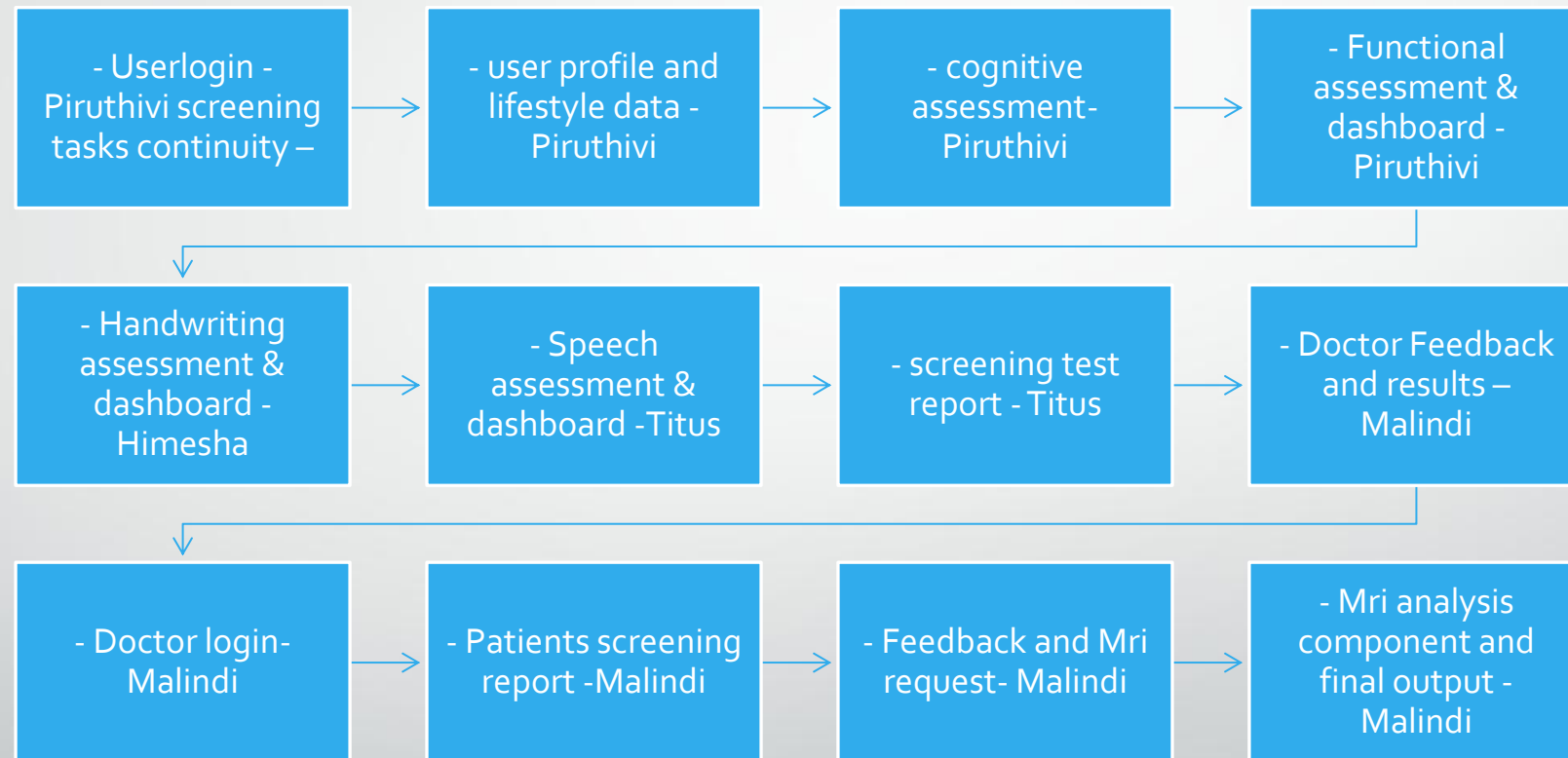
Translation to Real-World Deployment

- Pilot clinical validation study
- Model performance benchmarking (Accuracy, AUC, Sensitivity, Specificity)
- Regulatory & ethical compliance integration
- Integration into digital health ecosystems
- Multilingual deployment: English, Sinhala, Tamil

Backups -> AI-POWERD CONGNITIVE SCREENING AND ANALYSIS PLATFORM



Demo Plan



Thank You